

A full symmetric-group stabilizer on a tropical line

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Disproof of the maximum-stabilizer clause. The statement asserts that the largest stabilizer for the S_n permutation action on tropical cell structures and point orbits is S_{n-1} . The standard tropical hyperplane already has a point fixed by the whole S_n . We give the explicit $n = 3$ example, including the projective interpretation.

Work in the max-plus tropical semiring

$$\mathbb{T} = \mathbb{R} \cup \{-\infty\}, \quad a \oplus b = \max(a, b), \quad a \odot b = a + b.$$

The genuine tropical polynomial $F = X_0 \oplus X_1 \oplus X_2$ evaluates, on finite coordinates $x \in \mathbb{R}^3$, to

$$h(x) = \max(x_0, x_1, x_2).$$

Its tropical hypersurface is the corner locus

$$H = \{x : \text{at least two distinct coordinates attain } h(x)\}.$$

For a subset $S \subseteq \{0, 1, 2\}$, write

$$C_S = \{x : x_i = h(x) \iff i \in S\}.$$

The nonempty C_S with $|S| \geq 2$ are precisely the relatively open cells of H . In particular,

$$C_{\{0,1,2\}} = \{(c, c, c) : c \in \mathbb{R}\}.$$

This central cell is nonempty and contains $o = (0, 0, 0)$.

Define the coordinate action by $(\sigma \cdot x)_i = x_{\sigma^{-1}(i)}$. This is an actual action of S_3 , and permutation invariance of the maximum gives

$$h(\sigma \cdot x) = h(x), \quad \sigma \cdot C_S = C_{\sigma S}.$$

Thus it acts on the tropical hypersurface and its cell structure. For every $\sigma \in S_3$ we have $\sigma \cdot o = o$, so

$$\text{Stab}_{S_3}(o) = S_3, \quad |\text{Stab}_{S_3}(o)| = 3! = 6 > 2 = 2! = |S_2|.$$

The central cell also has setwise stabilizer S_3 . Since every stabilizer is a subgroup of S_3 , the full group is itself the largest stabilizer in this example, contradicting the proposed S_{n-1} maximum. In particular it is not merely a differently embedded copy of S_2 ; the groups have different orders.

Projective version. Tropical projective coordinates identify x with $x + c(1, 1, 1)$. The identity

$$h(x + c(1, 1, 1)) = h(x) + c$$

shows that H descends to this quotient, and the permutation action respects the equivalence relation. The central cell projects to the single point $[o]$, the central vertex of the standard tropical line. All six permutations fix $[o]$, whose actual projective stabilizer again has order six. The vector o has three finite tropical coordinates; it is not the excluded tropical zero vector $(-\infty, -\infty, -\infty)$.

Scope and verification. Both language versions make the largest-stabilizer assertion without excluding central cells, fixed points, or lower-dimensional cells. This disproof uses that unrestricted clause. It makes no assertion about a variant restricted to noncentral or maximal-dimensional cells, nor about the separate McKay-graph, orbit-count, or Young-subgroup clauses.

The Lean project defines F as an actual `MvPolynomial` over Mathlib's tropical semiring, proves its maximum evaluation, the complete corner-locus/cell description, permutation and translation invariance, and uses actual `MulAction.stabilizer` subgroups. It also constructs the quotient by common translations and its group action, proves that the central cell projects to a singleton, and computes both stabilizer cardinalities. All audited conclusions use only standard logical axioms. Reproduction instructions and pinned versions accompany the source and PDF.